

Sublexical modality

Fabienne Martin (Utrecht University)
f.e.martin@uu.nl

April 23, 2026

Second version of a chapter of the *Cambridge Handbook of Natural Language Modality*
(ed. by A. Arregui, V. Hacquard and M. Ippolito)

1 Modal causative verbs

Causal expressions are traditionally taken to describe causes and effects in the evaluation world. For instance, covert (lexical) causative verbs like *kill* or overt causatives like *cause* are generally assumed to describe actual results, not possible ones. But philosophers and linguists have often noticed that in English as well as other languages, some lexical causative verbs can be used in specific contexts to describe actions that fail to be causally successful in the actual world (Ryle 1949, Oehrle 1976). For instance, *Katja taught me Russian* or *Abi sent me her book* can be made true by completely unsuccessful attempts (perhaps I didn't learn any Russian or never received Abi's book).

Causative verbs like *teach* or *send* are special though in that they have been claimed to be 'sublexical modal verbs', that is, verbs that covertly encode a layer of modality while showing the morphosyntactic features of lexical causative verbs (Koenig & Davis 2001, Beavers 2010, Martin & Schäfer 2012). Sublexical modality is postulated to account for why these causative verbs are 'defeasible', that is, imply a result but do not entail it.

But result defeasibility also seems to be possible in an *ad hoc* (i.e. non-lexicalized) use with at least some causative verbs that Koenig & Davis (2001) call 'neutral', which entail successful causations in their normal use. To quote Ryle (1949):

[...] we very often borrow achievement [i.e. result] verbs to signify the performance of the corresponding task activities, where the hopes of success are good. [...] a doctor may boast that he is curing his patient's pneumonia, when his treatment does not in fact result in the anticipated recovery. 'Hear' is sometimes used as a synonym of 'listen', and 'mend' as a synonym of 'try to mend'. (Ryle 1949, 143)

This contextual flexibility speaks against a purely lexically restricted account of these 'failed attempt readings', on which the expected result is shifted to possible worlds.

Modal displacement with causative verbs gives rise to series of interrelated puzzles, which are addressed in turn in this chapter. First, there is the question of how to define modal causatives and the variation across the lexicon: what is the difference between verbs like *teach* or those like *cure*, and what is the source of Ryle's *ad hoc* failed attempt for the latter? Why is this *ad hoc* use easier for *cure* than for *kill*, for instance? A second puzzle is that it is easier to present the result as possible rather than actual when the subject of the lexical causative verb is animate and agentive. This point is illustrated in the examples below:

- (1) a. I offered him a spectacular view of Chimborazo through my binoculars. (But he declined.)
b. The flight over the Andes offered him a spectacular view of Chimborazo. (# But he missed it since he had pulled the shades down.) (Kratzer 2013a)
- (2) a. Pierre taught him Russian. But obviously, he did not learn anything.
b. This correspondence taught him Russian. #But obviously, he did not learn anything.

- (3) a. Bob tells me that winter is here, but I don't believe it.
 b. All this snow tells me that winter is there, #but I don't believe it. (Haïk 1989, 8)

The general pattern is that with animate and agentive subjects, verbs like *teach* can simply describe actions that typically lead to a certain outcome, but this outcome does not have to occur for the sentence to be true, as shown by the non-contradictory continuation in (1a)-(3a). With an inanimate subject, however, the same verbs often seem to entail the occurrence of at least part of the result, cf. the contradictory continuations in (1b)-(3b). The same pattern, which underlies Demirdache & Martin's (2015) Agent Control Hypothesis, has been shown to hold in many languages: Səncáəən (Kiyota 2008), Korean (Beavers & Lee 2020), Tagalog (Dell 1983, Alonso-Ovalle & Hsieh 2021), Kimagarang (Kroeger 2017), Malagasy (Paul, Ralalaoherivony & de Swart 2020), Abui (Kratochvíl & Delpada 2015), Totoli (i Farré, Riesberg & Himmelmann 2022) or Mandarin (Demirdache & Martin 2015).

However, a generalization in terms of animate versus inanimate subjects is only a simplification. For example, the occurrence of the outcome cannot be denied with any type of animate subject; in particular, the outcome cannot be denied anymore when the animate subject does not exercise agentive control over their own behavior (Davis, Matthewson & Rullmann 2009, Jacobs 2011, Fauconnier 2013, Beavers & Lee 2020). Conversely, the occurrence of the outcome can be denied without contradiction under certain conditions with inanimate subjects (Martin & Schäfer 2012, Gyarmathy & Altshuler 2020; see also Nederveen 2025 for related discussion), like in the presence of evidential adverbs such as *objectively* or *clearly* (cp. (4a)-(4b)).

- (4) a. This situation showed them the problem, #but they didn't see it.
 b. Clearly, this situation objectively showed them the problem! It is crazy that they didn't see it! (modelled after Martin & Schäfer 2012, 254)

A third question concerns the difference in the direction of the inferential path that often accompanies the distinction between the result-entailing use and the result-implying use (i.e. with or without result in the actual world). The pair below illustrates this:

- (5) a. The after sale service fixed the watch, but it still doesn't work.
 b. The impact fixed the watch. #But it still doesn't work.

In (5a), we envisage the causation event 'from left to right': we start with the repairman's action and an inference about the normal outcome of this action is made from there. This *prognosis* is subsequently invalidated. In (5b), however, we proceed in the opposite direction; we start with the outcome and make a *diagnosis* about the hypothesized cause through abductive reasoning (see Gyarmathy & Altshuler 2020, Vogelee & Guéron 2022). This difference in inferential directionality relates to the frequently reported intuition (see e.g. DeLancey 1981, Fauconnier 2012, Martin 2015) that causative sentences have a 'prospective', 'looking forward' flavour when the subject is a normal agent, while they have a 'retrospective', 'looking backward' flavour when the subject is an inadvertent agent or an accidental event: there is usually no awareness of accidents or uncontrollable outcomes until they have occurred. The last two puzzles are related: in (4b), *objectively* and *clearly* seem to induce the forward-looking perspective often not adopted by default with inanimate subjects.

Failed attempt uses of causative verbs constitute a subtype of a broader phenomenon, namely non-culminating construals, by which a telic predicate (that is, a predicate associating an endpoint to the events in its denotation) describes an incomplete event. This is surprising under the standard assumption that an untensed telic predicate denotes a set of complete events. In some languages, these constructions can appear not only with causative telic predicates, but also with non-causative ones, like *eat the cookie* (on the difference between causative and non-causative telic predicates, see Levin 1999, Rappaport Hovav 2008). For example, in Hindi, it is possible to say the equivalent of *Maya ate the cookie, but did not finish eating it* without contradiction, provided that the eating event has already begun (Singh 1991, Arunachalam & Kothari 2010, Altshuler 2014). These 'partial change' constructions are generally distinguished from 'failed attempt' ones, as they entail a partial change of the object's referent (see Tatevosov 2008, Demirdache & Martin 2015).

In some languages like Burmese (Kato 2014) or Tagalog (Santiago 2015), it is even possible to deny that the action proper has actually begun with (non-telic) activity verbs. In Burmese for example, Kato (2014) reports that failed attempt interpretations are possible with many activity verbs (e.g. *thàin* 'sit down', *pyé*

‘run’, *pyóun* ‘smile’, *ɕauʔ* ‘walk’). All verbs concerned are agentive. For instance, (6) is felicitous as soon as the agent minimally initiated the motor planning processes, even without successful motor execution (e.g., they tensed their muscles because they aim to walk, but could not properly walk because of pain).¹

- (6) *ɕauʔ=tê. dà=bêmê ɕauʔ=lô m̄-yâ=bú.* (BURMESE)
 walk=REAL this=though walk=LO NEG-get=NEG
 ‘(I) walked. But (I) couldn’t walk.’ (Kato 2014, 9)

One of the ways to account for these readings consists in shifting culmination or successful attempts from the base world to a set of modal alternatives, see discussion in 5.2 MODALITY AND ASPECT.² As illustrated in Figure 1, the modal operator responsible for incomplete interpretations may be located at different heights in the syntactic structure. From the bottom up:

- in the verbal phrase: (a) Koenig & Davis (2001) and subsequent work assume that some English verbal predicates (e.g. *send*) lexically specify a modal base relative to which event structure and argument structure are evaluated (see also section 2 below); (b) Koenig & Muansuwan (2000) propose that Thai change-of-state verbs encode a built-in imperfective operator (e.g., the Thai verbal stem *tɛɛŋ* is better translated as ‘be writing’ than ‘write’).
- in the Voice phrase, that is, in the projection headed by the functional head that introduces the external argument (the subject of transitive clauses). For instance, Bar-el, Davis & Matthewson 2005 assume a modal operator in the semantics of the control transitivizer in Salish languages, and Beavers & Lee 2020 make a similar proposal for a specific active Voice head in Korean.
- in the tense/aspect phrase: partitive tense-aspect morphologies (e.g. the English progressive) require only a part of an event in the extension of the VP that it combines with (Altshuler 2014, Martin & Gyarmathy 2019, see also 5.2 MODALITY AND ASPECT).

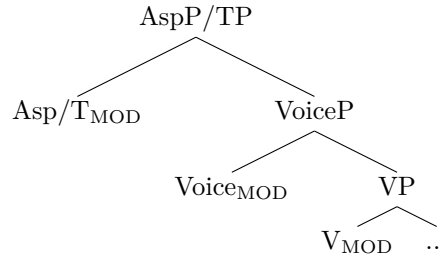


Figure 1: Possible positions of the modal operator underlying non-culminating interpretations of accomplishments

The most appropriate choice of where to locate the modal operator in the syntactic structure depends on the distributional restrictions that these interpretations are subject to in the language under consideration, in particular on whether they are confined to a specific class of verbs or instead tied to a specific Voice or tense-aspect morphology (see Koenig & Chief 2008, Koenig, Chief, Muansuwan & Shakti 2017, Martin 2019, Martin & Demirdache 2020 and Tatevosov 2020 for a cross-linguistic discussion of the advantages and drawbacks of each alternative). That said, agentive marking appears to facilitate a failed attempt interpretation across languages and verb classes, even when this use is licensed by an imperfective sitting in TP or AspP. For example, in English, a causative verb inflected in the progressive can have a failed attempt use in the context of an agentive subject only (Truswell 2011, Martin 2015, Harley & Folli 2023, Martin, Schäfer & Piñón 2025). This is illustrated in (7):

¹This makes failed attempt activities different from avertive or proximative constructions, which express that the event is narrowly avoided or almost happened and do not entail initiation (e.g. French *faillir P* ‘almost P’, English *be about to*).

²Competing, non-modal accounts of non-culminating accomplishments resort to a degree argument tracking the degree of event realization (Piñón 2008, 2009, Martin, Sun, Liu & Demirdache 2023) or the degree of change (Koenig & Chief 2008) in the semantics of accomplishments, or assume that the theme NP can denote proper or improper parts of the reference object (for instance, Soh & Kuo 2005 propose that Mandarin *yi-fen xin* ‘one letter’ can denote a part or a whole letter).

- (7) a. The noise of the dishwasher is waking up Ana, #but she’s dead to the world– she hasn’t even begun to flutter her eyes yet.
 b. I’m waking up Ana – I’ve shaken her twice! but she’s dead to the world – she hasn’t even begun to flutter her eyes yet. (modelled after Martin 2015)

Section 3 focuses on this generalization, namely that the subject must typically exhibit agency in order for the (whole) expected result to be relegated to possible worlds—what I will call ‘the agent requirement’. Before that, section 2 outlines the various treatments of low verbal modality (in the VP). Section 4 accounts for the agent requirement in an anchor-based semantics (see 1.3 THE CONSTRUCTION OF MODAL DOMAINS - THE KRATZERIAN VIEW). Coupled with Varasdi’s (2014) analysis of descriptions of incomplete events and Martin et al.’s (2025) semantics of lexical causatives, this analysis can account for the aforementioned puzzles in a more satisfactory way than previous accounts. Further, it will explain which property of agency is crucial for the failed attempt interpretation, based on a typology of atypical agents as they are grammaticalised across languages. The conclusion will be that *foreseeability* appears to be the essential agentive dimension. Section 5 concludes by examining the conditions under which causative verbs may give rise to defeasible readings. I suggest that such readings are facilitated by the low perceptibility of the target outcome, the surface indistinguishability of successful and unsuccessful action tokens, and the presence of manner information in the lexical semantics of the verb.

2 Sublexical modality

Koenig & Davis (2001) propose that many English verbs that are usually not considered as modal are in fact modal. They focus on English change-of-possession verbs (*give, deny, charge, offer*), verbs of interaction (*persuade, forbid, require, urge*) or perception (*perceive, miss, overlook*), but the key idea can in principle be applied to all types of verbs, such as Burmese activity verbs mentioned in the previous section. Their proposal is that verbs encode two types of information: situational information and modal information. The situational core (SIT-CORE) specifies the relations which characterize participants in the event. The sublexical modal component (a modal base) specifies restricted set of worlds and time indices at which the situational core is to be evaluated. This is illustrated in Figure 1, which gives the lexical frame they attribute to ditransitive verbs (i.e. verbs with two objects like *give, offer, charge*, etc).

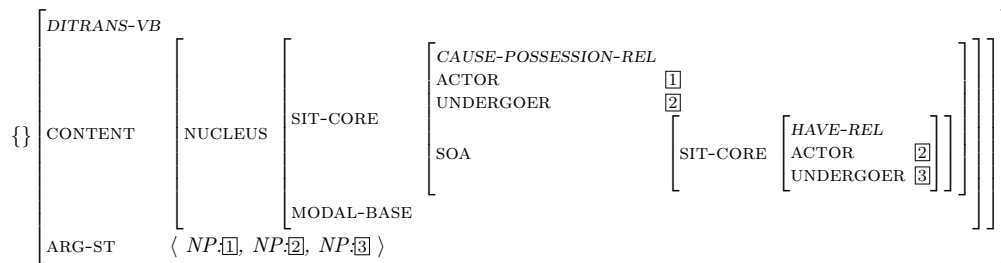


Figure 2: Lexical frame for ditransitive verbs (Koenig & Davis 2001, 106)

A key difference between sublexical modal verbs (e.g. *offer, force, miss*) and what Koenig & Davis (2001) relabel superlexical modal verbs (e.g. *must* or *can*) concerns the way the modal base is determined. With superlexical verbs, it is fixed contextually; for instance, *must* can have an epistemic, deontic, energetic or bouletic flavour depending on the context (Kratzer 1981). By contrast, for Koenig & Davis (2001), sublexical modals *lexically specify* a modal base. For example:

- (8) a. *send, offer, urge, defy*: energetic modal base
 b. *promise, charge, require, overlook*: deontic modal base

For energetic modals, the occurrence of the outcome is relegated to worlds in which ‘the actions or situations denoted by the verb achieve the goals that motivate those actions or the inherent consequences which result from the occurrence of the situation.’ (p. 89). The paraphrase (9b) of (9a) illustrates this idea.

- (9) a. Susan offered Brenda 10 euros.
 b. ‘Susan caused Brenda to have 10 euros in all worlds where the inherent consequences of her offer are achieved.’

Since linking constraints and event/argument structure depend only on the situational core component and are insensitive to the sublexical modality, they can explain why ‘neutral’ causatives such as *persuade* and those like *urge* have the same linking pattern, although only the former entail a result in the actual world (at least in their standard use). Koenig & Davis (2001) account for the difference between verbs like *persuade* and *urge* as follows: while the latter encode an irrealis (energetic, bouletic ...) modal base, the former represent a limiting case with a null modal restriction: the restricted set is identical to the entire set of circumstances in the initial world.

3 The agent requirement

Koenig & Davis (2001) only consider agentive subjects, so they do not address the fact that negating the outcome often yields a contradiction when the subject is non-agentive. Martin & Schäfer (2012, 2013, 2017) identified 55 verbs in French and German that accept both animate and inanimate subjects, show morphosyntactic features of lexical causative verbs, and are subject to the agent requirement (that is, the result is entailed or not, depending on whether the subject is an agent or not).³ Agent subjects (whether animate or inanimate; see section 4.2.3) denote an *individual* participating in the vP-event, whereas causer subjects denote *an event, a state or a fact* that brings about the vP-event (see Martin et al. 2025 for a detailed semantic characterization of these two types of external arguments). Martin & Schäfer (2012) assume that the sublexical modal has universal force and argue that the agent requirement can be accounted for through the way the type of subjects affects the choice of the modal base. With agent subjects, the modal base is typically energetic. Since the world of evaluation is not necessarily included in the energetic modal base (i.e., it is not necessarily a world in which the action performed by the subject achieves its inherent consequences), the result does not have to occur, wholly or partially, in the actual world. With causer (i.e. non-agentive) subjects as in examples in (1b-3b) and (4-5), they propose that the modal base is typically circumstantial: all worlds in the restricted modal base ρ share some facts with the base world. Since the base world is then obviously contained within the modal domain as it shares some facts with itself, it is not filtered out and necessarily quantified over (see also Alonso-Ovalle & Hsieh 2018).

To explain why, in some contexts like the one in (4b), the result entailment can be avoided with an inanimate subject, they propose that in the presence of a inanimate (causer) subject, these verbs can also be evaluated with respect to a stereotypical modal base. When, as in (4b), the context makes clear that the base world is an abnormal world, the result can then also be felicitously denied.

Martin & Schäfer (2012) analyse these verbs as denoting relations between an individual y , an event e , and a world w , such that e , whose theme is y , causes the result state s with y as theme in all worlds w in the modal base ρ accessible from w . This is illustrated in (10a/b), which gives the meaning for the VP *offrir y à z* , adopting Alonso-Ovalle & Hsieh’s (2018) modifications of Martin & Schäfer’s (2012) lexical representations. The two conditions associated with the verb in (10b) ensure that the existence of an agent implies an energetic modal base, while the existence of a causer implies a circumstantial or a stereotypical modal base.

- (10) a. $[\text{VP offrir } y \text{ à } z] \rightsquigarrow$
 $\lambda y \lambda z \lambda e \lambda w [\text{offer}_w(e) \wedge \text{theme}(e, y) \wedge \text{recipient}(e, z) \wedge$
 $\Box_{\rho_w}^{w'} \exists s (\text{cause}_{w'}(e, s) \wedge \text{have}_{w'}(s) \wedge \text{possessee}(s, y) \wedge \text{possessor}(s, z))]$
 $=_{\text{def}} \lambda y \lambda z \lambda e \lambda w [\text{OFFER}(\rho_w, e, z, y)]$
- b. Conditions:
- (i) $\forall e \forall z \forall y \forall w (\text{OFFER}(\rho_w, e, z, y) \wedge \exists x (\text{agent}(e, x)) \rightarrow$
 $\rho_w = \text{energetic})$
 (the existence of an agent implies an energetic modal base)
 - (ii) $\forall e \forall z \forall y \forall w (\text{OFFER}(\rho_w, e, z, y) \wedge \exists x (\text{causer}(e, x)) \rightarrow$

³The complete list of verbs is available on zenodo.org/records/19604573

$\rho_w = \text{circumstantial} \vee \rho_w = \text{stereotypical}$
 (the existence of a causer implies a circumstantial or a stereotypical modal base)

In (10a), a (manner) property is ascribed to the verbal event. This is necessary to ensure that the (possibly unsuccessful) event is an offer. For instance, when John Kennedy bought and packed the present he aimed to offer to his wife during their stay in Dallas in 1963 and got assassinated before being able to present it as a gift to her, the buying and the packing doesn't make the sentence *John offered her his present* true, no matter whether his offer would have been successful or not. Encoding an event property is not typical of causative verbs though, since the latter are often not supposed to say anything about the way the change is brought about.⁴

While Martin & Schäfer's (2012) proposal seems to account for the agent requirement, it remains unsatisfactory for several reasons. Firstly, as pointed out by Piñón (2014), it amounts to assuming that defeasible causative verbs are lexically ambiguous, since the choice of the modal base depends on the choice between an agent or causer subject. The ambiguity is not even a case of regular polysemy, since there is no systematic semantic relation between agent and causer subjects. Secondly, one would like to understand *why* the whole outcome can be denied in the context of an agent subject but not in the context of a causer subject. Furthermore, it is unclear why causative verbs exhibiting the agent requirement should only instantiate the three modal flavours mentioned in the two conditions. Some of these verbs have been attributed other modal flavours. For instance, French *promettre* has a deontic reading with an agent subject discussed in Koenig & Davis (2001), and an epistemic reading with an inanimate subject (Ruwet 1991, see also Vázquez-Laslop 2000 on Spanish *prometer* 'promise'). For French *permettre* 'allow/enable', Hacquard (2006) distinguishes a real deontic ('grant permission') reading with an agent subject, and an enabling reading with an inanimate (causer) subject. Only the latter entails the occurrence of the result, but these two values for ρ do not appear in the conditions in (10). Furthermore, given that *permettre* 'allow/enable' has an abilitative meaning, it is debatable whether the modal has universal rather than existential force. As seen above, however, Martin & Schäfer's (2012) account of the result entailment with causer subjects relies crucially on the assumption that the sublexical modal has universal force.

Another issue is that the value for ρ does not appear to depend solely on the verb and the type of external argument. Consider, for example, (11):

- (11) Fate obliged me to move.

Depending on how the speaker conceives fate, the modal domain will be either (theo-)bouletic (a subcategory of energetic modality), (theo-)deontic, or circumstantial. However, (11) entails that I moved regardless of the modal restriction (except perhaps if fate is personified). With an animate agent, however, denying the outcome would still be possible, as we do not always fulfil our obligations.

Furthermore, a clearly abnormal context does not necessarily suffice to avoid the result entailment with inanimate subjects, although it does invite a stereotypical restriction for the modal; compare (4a/b) with (12a/b). Note that (12b) is strange independently of the result denial.⁵

- (12) a. The impact fixed the watch. #But it still doesn't work.
 b. #Clearly, the impact objectively fixed the watch! It's crazy that it still doesn't work!

A further issue concerns undefeasible (neutral) causative verbs like *persuade* or *kill*. If we wish to retain Koenig and Davis's appealing idea that all causative verbs are modal, the presence of an agent cannot be sufficient to select an irrealis (energetic, bouletic...) modality. Circumstantial modality should then be lexically hard-wired into these verbs, so as to prevent them from having a failed attempt reading in the presence of an agentive subject.

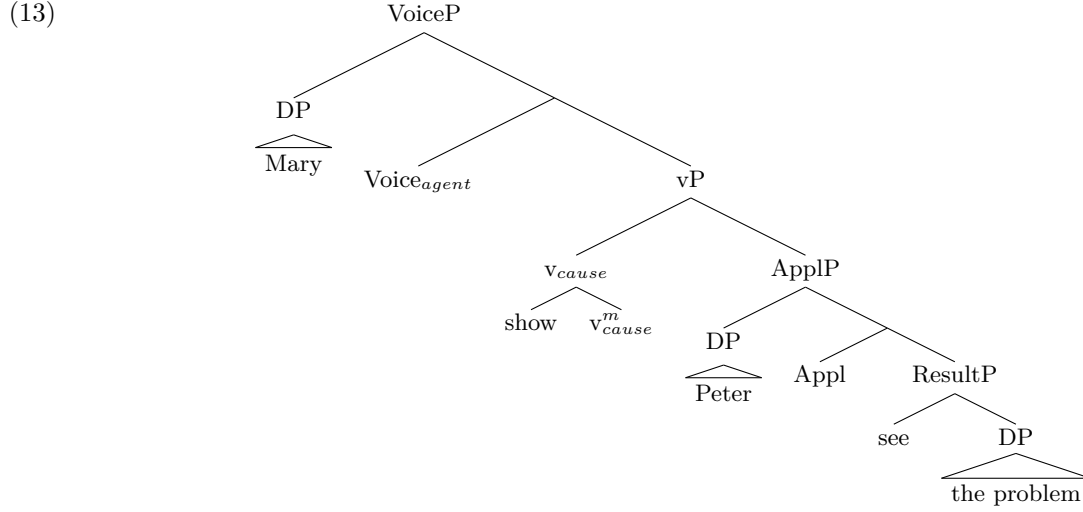
A last problem with the approach developed by Koenig & Davis (2001) and pursued by Martin & Schäfer (2012) is that there is no explanation as to why, in English, only causative verbs give rise to failed attempt

⁴Another atypical property of these verbs is that they do not participate to the causative/anticausative alternation (cf. *John broke the vase/the vase broke*). This is true of all 55 verbs of Martin & Schäfer's (2012) list.

⁵As a reviewer astutely observes, (12b) improves if some mechanical component slightly misaligned gets realigned due to the shock, and the speaker assumes that this realignment should objectively have led to a repair of the watch, an intuition accounted for in section 4.2.3.

readings, unlike in languages such as Burmese, for example, where the normal course of events can be pushed to possible worlds even with activity verbs (recall (6)). A related problem is that these approaches do not identify the element that introduces the modality in the verbal structure.

Assuming with Koenig & Davis (2001) that neutral causative verbs are also modal, a first option consists in having a modal operator introduced by little v_{cause} , the verbalising functional head that introduces the verbal event in a causative verb (Kratzer 1996, Harley 2012, Alexiadou, Anagnostopoulou & Schäfer 2015). This option is illustrated in (13) for the sentence *Mary showed Peter the problem*, implementing Martin & Schäfer’s (2012) idea that *offer* predicates a property on the verbal (causing) event) (in v_{cause}^m , ‘ m ’ indicates the modal meaning of the v head). The indirect object is introduced by an applicative head (see e.g. Bruening 2021).⁶



We then have modal quantification over a causal relation, CAUSE keeping its usual result-entailing, non-modal semantics. This idea can be implemented by replacing the ‘regular’ v_{cause} -head with the semantics in (14a) with a modal version of it with the semantics in (14b) (where v is a variable ranging over events or states, i.e. eventualities).

- (14) a. $v_{cause} \rightsquigarrow \lambda P \lambda e. \exists v (\text{cause}(e, v) \wedge P(v))$
b. $v_{cause}^m \rightsquigarrow \lambda P \lambda e \lambda w. \Box_{\rho_w}^{w'} \exists v (\text{cause}_{w'}(e, v) \wedge P_{w'}(v))$ (to be revised)

A second possible option is to give up the assumption that CAUSE necessarily entails a result in the evaluation world (when unembedded under a modal); see Copley & Wolff (2014), Copley & Harley (2015) and references therein. For instance, Hill (2024) argues that overt causatives are circumstantial modals. The presence vs. absence of result entailment can then be accounted for via the difference between modal necessity and possibility.

To account for why failed attempt readings are available for activity (non-causative) verbs in Burmese but not in English, one could posit a modal counterpart of little v_{do} , the verbalizing head that introduces the verbal event in the structure of activity verbs (Harley 2012). While in English, activity verbs can presumably only be built with the extensional version of v_{do} , Burmese resorts to its modalized version. In this language, activity verbs are in this perspective another kind of sublexical modal verbs.

In the next section, I show how the agent requirement and the other puzzles raised by failed attempt uses of lexical causatives can be solved adopting the anchor semantics for modals developed in Kratzer (2013b).

⁶See Bruening (2021) for an alternative analysis according to which modality of defeasible causatives is introduced by the applicative head Appl in the structure of ditransitive verbs. One problem with this view is that it is not generalizable to defeasible causative verbs that are not ditransitives, like e.g. *flatter* ‘flatter’, etc. Among Martin & Schäfer’s (2012) list of 55 defeasible causative verbs, 22 only are ditransitive verbs.

4 Defeasible causatives in an anchor semantics

As the previous section shows, it is a bit unclear whether the modal base for lexical causative verbs is fully lexicalized rather than contextually determined as it is with overt modal verbs (Hacquard 2006, Kratzer 2013b, Phillips & Kratzer 2024; see also Hill 2024 on the flavor variation of overt causative verbs). Beyond the subject’s type, the verb itself certainly contributes to determining the modal base, but this is probably just one of many factors influencing the interpretation of the modal. All in all, the reviewed data suggests that the agent requirement perhaps cannot be explained mechanically in terms of a distinction between types of external arguments and the respective values for ρ .

Elaborating on Arregui (2005) and Hacquard (2006), Kratzer (2013b) and Phillips & Kratzer (2024) propose that the modal base of modal expressions is built on the basis of two ingredients. Firstly, the *anchor situation* is a part of the actual world which serves as the ‘starting block’ for the modal thought reported in a modal statement. Secondly, from this anchor situation, one considers a set of possible extensions (or modal alternatives), that all have an exact counterpart of this anchor situation. This set of possible extensions forms the modal domain. From this huge set of possible extensions, many will not be considered at all; we therefore restrict this set further. Normality guides default restrictions for modal domains (Phillips & Kratzer 2024); different kinds of restrictions yield different modal flavours.

Grammar constrains the choice of the anchor. Kratzer (2013b) proposes that the anchor of a modal expression is one of its arguments. So, what is the anchor of defeasible causatives? A natural candidate is the event argument. To understand what the real-world counterpart of the modal anchor of defeasible causatives is, we must first identify the types of events which they refer to.

4.1 The modal anchor of causative verbs

Defeasible causative verbs have just the same argument and event structure as run-of-the-mill causative verbs (Koenig & Davis 2001). Therefore, the discussion about normal (neutral) lexical causative verbs on these particular issues can be extended to defeasible causatives.

Many authors assume that causative verbs have the same event structure when used as agentive or non-agentive transitives (*John/The wind opened the door*), or as intransitives (*The door opened*): in all uses, they denote a set of events that yield some state satisfying the stative predicate encoded by the verb (hereafter referred to as a ‘*P*-state’); see Rapp & von Stechow (1999), Schäfer (2008), Alexiadou et al. (2015). There is disagreement about the nature of the event-state relation. For Alexiadou et al. (2015), causative verbs are decomposed into v_{cause} and a stative predicate in both their transitive and intransitive use (see *John broke the vase/The vase broke*). The v_{cause} head introduces a CAUSE event that leads to some *P*-state. For Rapp & von Stechow (1999), these verbs are decomposed into v_{become} and a stative predicate. This head introduces a BECOME event developing into some *P*-state. Beyond this disagreement about the nature of the event-state relation, a shared assumption among these authors is that change-of-state verbs encode the *same* event-state relation across all uses.

Based on a variety of arguments related to the interpretation of adverbs, the progressive and aspectual verbs in combination with causative verbs, Martin et al. (2025) abandon this last assumption and propose that the event-state relation is interpreted differently across grammatical contexts. *Only* in the context of an agent subject, the verbal event is a CAUSE event. In all other contexts, the verbal event is a mere change (a BECOME event). Therefore, a syntax/semantics mismatch characterizes non-agentive causative verbs: they are syntactically transitive, but they have the same intransitive (inchoative) semantics as change-of-state verbs in their intransitive use. More concretely, in *Charlotte killed Jean-Paul*, the killing event is identified with a causing event, as Charlotte is an agent. But in *The accident killed Jean-Paul*, the killing event is identified with a change only (the dying event). The ‘causer’ subject (here: the accident) does not denote a participant to the verbal event; rather, it denotes its cause. One of Martin et al.’s (2025) arguments involves temporal adverbs:

- (15) a. Charlotte pushed Jean-Paul out of the window **on Sunday** and he fell from the third floor. #She ended up killing him **on Tuesday night**.
- b. Jean-Paul fell **on Sunday** from the third floor. After much agony and medical attempts to save him, the fall ended up killing him **on Tuesday night**.

Temporal adverbials such as *on Tuesday night* locate the verbal (here killing) event. Since (15a) is construed with an agent subject, the killing event is, by assumption, a causing event. Since this event is asserted to happen on Sunday, the continuation asserting that it happens on Tuesday night generates a contradiction (Fodor 1970).⁷ But in (15b), the subject is a causer, which introduces an event on its own (the fall), *different* from the killing event e , and causing e . The killing event is, by assumption, just a change (BECOME event). It can therefore be located on Tuesday night, even if it is taken for granted in the context that the fall took place on Sunday.

The same argument can be extended to defeasible causatives. For instance, in (16a), with an agent subject, the showing event is a causing event performed by the professor. This event is asserted to happen today, and therefore, the professor’s action must also happen today. This makes the information in parentheses contradictory. However, in (16b), with a causer subject, the showing event is the (mental) change which the theme is undergoing. There is no contradiction in locating the *cause* of the verbal event (the past lesson) at a previous point in time.

- (16) a. Our professor (#who I saw for the last time **last year**) showed me the solution **today**.
b. I never understood why our professor emphasized that particular analysis **last year** until I encountered this research problem. That past lesson showed me the solution **today**.

This means that, at the level of the verb and the vP, the relation between the verbal event and the state is not determined yet. It is only specified once the vP is embedded in a larger structure, containing a voice head, for instance. To account for this, Martin et al. (2025) assume that rather than introducing *cause* or *become*, the verbalizing head that enters the formation of change-of-state (cos) verbs comes with a relational argument R between events and eventualities, as shown in (17).

$$(17) \quad v_{cos} \rightsquigarrow \lambda P \lambda R \lambda e. \exists v (R(e, v) \wedge P(v))$$

The choice of Voice head determines how the event-state relation in the vP is interpreted. The combination of the Agent Voice head with a change-of-state vP translates R as *cause*, while the combination of a change-of-state vP with the head Causer Voice translates R as *become*.

I adopt this proposal and add to it the idea introduced in the previous section that the verbalizing head entering the structure of all⁸ causative (change-of-state) verbs is responsible for introducing the modal operator, see (18):

$$(18) \quad v_{cos}^m \rightsquigarrow \lambda P \lambda R \lambda e \lambda w. \Box_{\rho_w}^{w'} \exists v (R_{w'}(e, v) \wedge P_{w'}(v))$$

A predicate such as *show Abi the solution* can be attributed the meaning in (19) (the first object is introduced by an applicative head that shares its arguments with the verb, see Bruening 2021 and references therein). Again, an event property is predicated over the verbal event in order to ensure that it is a showing event (rather than a disclosure, for instance). The combination of this predicate with an external argument via Agent Voice or Causer Voice is given in (20).⁹

$$(19) \quad [\text{show Abi the solution}] \rightsquigarrow \\
\lambda R \lambda e \lambda w. \text{show}_w(e) \wedge \text{recipient}(e, \text{abi}) \wedge \text{theme}(e, \text{the-solution}) \wedge \\
\Box_{\rho_w}^{w'} \exists s (R_{w'}(e, s) \wedge \text{see}_{w'}(s) \wedge \text{theme}(s, \text{the-solution}) \wedge \text{experiencer}(s, \text{abi}))$$

⁷Like Alexiadou et al. (2015) and Rapp & von Stechow (1999), Martin et al. (2025) assume that change-of-states only have one event argument. Therefore, even if the verbal event contains two subparts (the action+the change-of-state), these subparts are not accessible for separate temporal modification.

⁸I adopt here Koenig & Davis’s (2001) suggestion that neutral causatives (e.g. *persuade* or *kill*) are also modal. These causatives must then be treated as sublexical verbs that lexically encode circumstantial modality. Another solution is to assume two different v heads in the formation of English change-of-state verbs, a modal and a non-modal one, such that defeasible causative verbs pick up the the first, and the neutral causative verbs the latter. Under the first option, *ad hoc* defeasible uses *à la* Ryle can be treated as cases in which circumstantial modality is overridden by irrealis modality under contextual pressure. Under the second option, these *ad hoc* use amounts to a genuine additional sense of the verb.

⁹See Martin et al. (2025) for more technical details. In (20b), *cause*⁺ is a relation between an event (denoted by the verb) and a state, and in (20d), *cause* relates a situation (denoted by the causer subject) and an event (denoted by the verb).

- (20) a. The professor showed Abi the solution.
 b. $\lambda e \lambda w. \text{agent}(e, \text{the-professor}) \wedge \text{show}_w(e) \wedge \text{recipient}(e, \text{abi}) \wedge \text{theme}(e, \text{the-solution}) \wedge$
 $\Box_{\rho_w}^{w'} \exists s (\text{cause}_{w'}^+(e, s) \wedge \text{see}_{w'}(s) \wedge \text{theme}(s, \text{the-solution}) \wedge \text{experiencer}(s, \text{abi}))$
 c. That past lesson showed Abi the solution.
 d. $\lambda e \lambda w. \text{cause}_w(\text{that-past-lesson}, e) \wedge \text{show}_w(e) \wedge \text{recipient}(e, \text{abi}) \wedge \text{theme}(e, \text{the-solution}) \wedge$
 $\Box_{\rho_w}^{w'} \exists s (\text{become}_{w'}(e, s) \wedge \text{see}_{w'}(s) \wedge \text{theme}(s, \text{the-solution}) \wedge \text{experiencer}(s, \text{abi}))$

When the predicate in (20b/d) combines with past tense, the verbal event e gets existentially quantified over. Applied to an event predicate, the English past tense returns a complete event. Thus, (20a) and (20c) describe a complete showing event in the evaluation world, which, *in the modal extensions*, causes (in (20a)) or develops into/becomes (in (20c)) a state of seeing the problem experienced by Abi.¹⁰

We can now return to our initial question: what does the modal anchor of defeasible causatives (the event argument) correspond to in the real world? In our example, in the context of an agent, the anchor (the verbal event e quantified over by Tense) is an action involving three participants, the agent, the recipient and the theme. (This event develops into an event causing a state of the right kind *in the modal extensions* only). As a recipient of this agentive showing event, Abi is not necessarily affected, thus the verbal event does not need to include the target change, at least in order to meet the thematic restrictions.

In the context of a causer, things are different: the anchor (the verbal event) only involves Abi and the solution as participants. The past lesson denoted by the subject is not a participant of the verbal event, but the cause of it ('causer' is not a theta-role). The anchor (the verbal event) is therefore now an *internal* showing event taking place within Abi's mind. What could this event be, if not Abi's discovery of the solution? Similarly in (1)-(4), the verbal event serving as modal anchor is identified with the formation of a perception of a spectacular view of Chimborazo in (1b), the learning of Russian in (2b), the formation of the belief that winter is here in (3b), and the discovery of the danger in (4a). With a causer subject, the anchor event is therefore identified with all or part of the change-of-state, and the modal is trivialized since its prejacent is already true in w .

4.2 Explaining the puzzles

We are now ready to revisit the puzzles mentioned in the introduction, starting with the way the subject influences the interpretation of defeasible causatives.

4.2.1 Revisiting the agent requirement

Obviously, the result entailment can only be prevented if the anchor event does *not* include all or part of a change-of-state of the relevant type endured by the internal argument. For if the change-of-state (or part of it) is included in the modal anchor, its occurrence will be assumed in the base world as this event will be quantified over by tense/aspect. Therefore, an agent subject makes it easier to avoid the result entailment, as this is the only type of subject that allows the anchor event to be 'change-free'. We capture the agent requirement, and this without making any assumptions on the modal restriction at play, or having to assume that defeasible causatives are lexically ambiguous (recall the problems raised by Martin & Schäfer's 2012 approach raised in section 3). We do not even have to assume that the sublexical modal has universal force to achieve this, what enables us to cover cases such as *permettre* 'allow/enable', where the modal probably has existential force. What we still need are two different Voice heads, but these were required in the previous framework anyway.

What other event bit(s) should the anchor event include as its subparts apart from the action performed by the agent? This question is in fact very similar to the one raised by progressive sentences: we must know 'how big' the anchor event must be in order not only to satisfy the thematic and lexical restrictions imposed by the verb on the verbal event discussed in section 4.1, but also to qualify as a 'causing event in the making'.

To answer this question, we can apply Varasdi's (2014) theory on the progressive to modal causatives. Varasdi (2014) builds on Asher's (1993) idea that the progressive is true of an event e by virtue of a certain set of properties F that the event e has. Varasdi's (2014) proposal is that the progressive refers to certain

¹⁰If the predicate combines with an imperfective such as the English progressive, the sentences describe a part of a showing event with the same properties.

Type of agents	priori intention to ϕ	full control	foreseeability	failed attempt
full agent	✓	✓	✓	✓
mistaken intentional	✓ (de re only)	✓	✓	✓
absent-minded	✗	✓	✓	✓
purely inadvertent	✗	✗/✓	✗	✗
circumstantially-enabled	✗/✓	✗	✗	✗

Table 1: Subtypes of atypical agents and the failed attempt reading (‘ ϕ ’ symbolises the description of the action by the verbal predicate, ‘✓’ and ‘✗’ in the last column signal the availability/unavailability of the failed attempt reading)

necessary conditions for the culmination. The relevant set of necessary conditions F divides into two subsets. Indicative conditions have to do with properties of the ongoing event that single out the relevant event type with respect to the set of contextual alternatives; informally, these conditions correspond to the informational cues that enable us to recognize that the event e is an ‘event of the relevant type in the making’. Sustaining conditions are required for the event to make progress toward the indicated outcome.

Returning to modal causatives, we can similarly assume that the anchor event e of defeasible causatives must involve as many informational cues and circumstances of the base world as are necessary for it to qualify as a causing event in progress.

Is it sufficient for the modal anchor to involve an action performed by the agent for it to qualify as such? It certainly is the case when we deal with a ‘normal’ agent, that is, an agent that is in-control of their behaviour and intends to bring about an event that fits the description of the verbal predicate, since intentional and in-control actions are highly indicative of their goal (Martin 2015), but see section 5 for a more nuanced generalisation.

4.2.2 Atypical agents

Whether the result entailment can still be avoided if we turn to abnormal or atypical agents very much depends on the kind of inadvertent agents we deal with. The generalization is that the result inference is still defeasible as long as the anchor event remains indicative enough of the outcome despite the reduced agency of the subject’s referent. The result inference becomes an entailment when the outcome is *unforeseeable* before the causing event culminates. This is somewhat the hallmark of accidental events: they are not seen beforehand and are typically cognitively apprehended *post facto* (DeLancey 1981, 649, Fauconnier 2012, 131; see also Dell 1983, 192).

A survey of the literature (Searle 1983, Inman 1994, Kamp 1999-2007, Kittilä 2005, Ganenkov, Maisak & Merdanova 2009, Kittilä 2013 a.o.) reveals four types of atypical agents as they are grammaticalized across languages, two of which allow for a failed attempt reading of causative verbs (see Table 1).¹¹

In the case of a *mistaken intentional agent*, the event description ϕ given by the VP corresponds to the agent’s intention taken *de re* only, as the agent does not have the *de dicto* intention to perform the action described by ϕ . Imagine for instance that Abi wants to save space on her hard disk and is mistaken about the content of the particular file she deleted, namely her dissertation draft. She may have the *de re* intention to delete her dissertation draft, but she certainly does not have the corresponding *de dicto* intention.

With a mistaken intentional agent, the failed attempt reading is possible (Martin & Schäfer 2012; see also Oehrle 1976, 84, Beavers & Lee 2020, section 3.3). Let’s consider Abi’s case again. We all go through frustrating episodes with computers where we do everything that should lead to a certain result, but the result doesn’t happen. Imagine how annoyed she is as she tries unsuccessfully to delete the file she fails to correctly identify as her dissertation file, shouting something like (21):

- (21) *Mistaken intentional agent*
But I DELETED it! Why is it still there?

¹¹English is taken as an illustrative language here; see Martin (2016) for cross-linguistic data.

A second type of inadvertent agents that license the failed attempt reading is the *absent-minded agent*, who do things routinely, automatically, and without premeditation. Since at least Searle (1983), philosophers have commonly distinguished between two types of intentions (Brand 1984, Pacherie 2015). The *prior intention* to ϕ is the belief-desire complex that prompts the agent to perform an action of type ϕ . The *intention-in-action* $_{\phi}$ coordinates the agent’s activities during the unfolding of the ϕ -action.

Searle (1983) proposes that a ϕ -action can be controlled by the agent’s intention-in-action, without being prompted by a prior intention to perform the action ϕ . He provides the example of a philosopher in a chair, lost in thought, who suddenly gets up and starts pacing the room, without being prompted by a prior intention to do so.

In English, some adverbials convey ‘absent-minded agency’, such as *absent-mindedly*, *spontaneously*, *without really wanting to*, and so on. Some languages have a dedicated morpheme to express this type of atypical agents. For instance, the suffix for the spontaneous Voice in classical Japanese signals that the agent does not intend to bring about an action, but nevertheless does so under the pressure of some agent-external circumstances (Shibatani 2006, 223).

The failed attempt reading of causative verbs is possible with absent-minded agents. For instance, assuming that Abi absent-mindedly does exactly what one is supposed to do to delete a file, but without the prior intention to do so, the failed attempt reading is felicitous in (22).

(22) *Absent-minded agent*

Abi absent-mindedly deleted the file. But very luckily the manoeuvre wasn’t successful!

With absent-minded and mistaken intentional agents, the action is extensionally undistinguishable from the action performed by a normal (intentional, in-control) agent and the eventual outcome is already identifiable before the action has actually brought it about. The action can therefore be as indicative of the potential outcome as when the subject is a prototypical agent. The verbal (anchor) event can therefore just involve this action and still qualify as an ongoing causing event. The normal outcome of this action can then be relegated to possible extensions of this anchor.

The two remaining types of atypical agents block the failed attempt reading (see Beavers & Lee 2020 on Korean, Kiyota 2008 on Sanchaean, Tsujimura 2003, 397-398 on Japanese, Dell 1983, Alonso-Ovalle & Hsieh 2021 on Tagalog, Kroeger 2017 on Kimaragang, Paul et al. 2020 on Malagasy, i Farré et al. 2022 on Totoli). The first type is the *purely inadvertent agent*. With such agents, the event description ϕ given by the VP does not correspond to neither the agent’s *de dicto* prior intention, nor to their *de re* prior intention. Furthermore, this agent either lacks agentive control over their behavior or, if they do exercise such control, nevertheless fails to foresee the consequences of their action in advance. In both cases, the outcome of the action is not foreseeable prior to its actual realization. For instance, imagine that Abi is either cleaning her computer’s keyboard (Abi exercises agentive control) or falling on it because she stumbled (Abi does not exercise agentive control). While doing so, she *unexpectedly* deletes her dissertation file. In this context, denying the result is rather infelicitous, see (23).¹²

(23) *Purely inadvertent agent*

While she was falling on/cleaning the keyboard, Abi unexpectedly/inadvertently deleted the file. #But very luckily, the file is still there/the manoeuvre wasn’t successful!

The fourth and last type of atypical agent is interesting because it can be an intentional agent. A number of languages have a particular verbal morphology called limited-control or out-of-control (see Bar-el 2005, Bar-el et al. 2005, Jacobs 2011 on Salish, Travis 2010, Paul et al. 2020 on the *-aha* construction in Malagasy; Alonso-Ovalle & Hsieh 2021 on the ability/involuntary action form in Tagalog). When these morphologies are used in contexts that make clear that the agent is intentional, they typically convey that the agent is not in full control of what they are doing, and achieves their goal only with the help of external circumstances (luck, fate, etc.): ‘someone who has to struggle [...] lacks full control.’ (Delancey 1985, 393). Informants often translate sentences involving this kind of ‘circumstantially-enabled’ agents with English *manage to*.

¹²Kamp (1999-2007) conjectures that adverbials signalling atypical agent pick up different types of atypical agency. It seems that while the adverbials *accidentally* and *inadvertently* can be used for both inadvertent readings, ‘*by mistake*’ preferably selects the ‘mistaken intentional’ agent. See also Beavers & Lee (2020) on the difference in Korean between *silswulo* ‘mistakenly’ and *wuyenhi* ‘accidentally’.

Across languages, out-of-control morphologies block the failed attempt reading of causative verbs. For instance, in Halkomelem, the failed attempt reading that is available with the control transitivizer in (24a) is not felicitous in the presence of the limited-control transitivizer (data from Gerdtz 2008, cited by Jacobs 2011, 25).

- (24) a. *control predicate (in-control agent)* (SALISH)
 niŋ cən qa:y-t t^φə speŋəə ŋiŋ ŋəwə niŋ-əs qay.
 AUX 1S.SUB kill-**tr** DET bear and NEG AUX-3SSUB die
 ‘I killed the bear but it didn’t die.’
 b. *limited-control predicate (circumstantially-enabled agent)*
 #niŋ cən qəy-nəx^w t^φə speŋəə ŋiŋ ŋəwə niŋ-əs qay.
 AUX 1S.SUB kill-**lctr** DET bear and NEG AUX-3SSUB die
 ‘I managed to kill the bear but it didn’t die.’

With purely inadvertent and circumstantially-enabled agents, the potential outcome cannot be foreseen on the basis of the action alone. The event anchoring the modal must therefore include, beyond the action itself, a richer set of background circumstances drawn from the base world to qualify as a causing event of the right type in the making. This leads to a dead end for the failed attempt reading.

Consider, for example, Abi’s cleaning of the computer or falling on it in (23). These actions do not indicate the ‘delete-dissertation-file’ event type — her action isn’t identifiable as such *before* she unfortunately triggers it. The anchor event must therefore include (some of) the change for it to qualify as a deleting event in progress.

In other cases, the anchoring event must encompass more circumstances of the base world than just the subject’s action to fulfill enough necessary conditions for the right type of causation, e.g. the help of fate, luck, or some other additional circumstances. For instance, considered in isolation, my non-expert attempt to kill a bear certainly does not qualify yet as a killing-a-bear in progress (in fact, considered on its own, my novice attempt is rather *anti*-indicative of success!). External enabling forces have to be part of the anchor too for it to indicate success. However, once these external forces have been incorporated into the anchor event, the latter inevitably *also* leads to the change, and this either in the base world or in all modal extensions.

There are two causes for this, which may be concurrent. Firstly, it may be impossible for these circumstances to occur without the change also occurring: these circumstances are either consubstantial with the change (e.g. what luck or fate does *is* trigger the change), or they “conspire to make things happen” (Weera-sooriya, Rivero & Arregui 2022, see also Inman 1994 on the modality of inevitability). Secondly, it may be that the causal role of these circumstances can only be identified once the change has begun. Sometimes we indeed realise that external forces have helped us to achieve a particular outcome only once we have achieved it. But then, conveying (via e.g. an out-of-control morphology) that the outcome was achievable only with the help of these external forces *de facto* amounts to conveying that the outcome *has* been achieved.

The latter situation is typical of the crossover between the causal time and the evidential time which can be observed in reports of inadvertent events.¹³ The causal time axis orders events in the real world; on this axis, the out-of-control action (e.g. Abi’s unexpected manipulation of the keyboard) precedes the outcome (the disappearance of the file). The evidential time axis organizes how evidence becomes available over time. Typically, we gain epistemic access to the outcome before we gain access to the inadvertent action causing it; in fact, discovering the outcome often leads to the cognitive apprehension of the inadvertent action. It is because of this crossover between causal time and evidential time that we often get the impression that the temporal perspective is retrospective in causative sentences involving an accidental agent (Vogeleer & Guéron 2022). On the evidential axis, we indeed typically start from the result and work our way back to the cause. With normal agents, however, events and their respective cognitive apprehension follow the same order on the causal and evidential axes.

¹³See Merin (2003, 2004) on the distinction between causal and evidential time, as well as Martin (2018, section 3.4) and Nadathur & Lauer (2020, section 3.5) for possible ways to define the relation between the causal time and the evidential time in the semantics.

4.2.3 Inanimate subjects

Since inanimate subjects are preferentially construed as causer subjects (Folli & Harley 2005, Alexiadou et al. 2015), a result entailment is typically triggered with such subjects (see section 4.2.1).

However, Folli & Harley (2008), Fauconnier (2012), Martin et al. (2025) and others insist that grammatical agents can be inanimate. For Martin et al. (2025), the distinction between agent and causer subjects is not based on animacy, but rather on the semantic type of the external argument: agents are individuals, causers are eventualities, facts or other kind of abstract objects. Therefore, depending on how it is construed, an inanimate subject is an agent or a causer in the grammar.

The result entailment is therefore also avoidable in principle with inanimate subjects, provided they are construed as agents, because in this case, the anchor event can be ‘change-free’.

Since the agent role is not the natural role for inanimates, the context must sometimes be modulated in order to favour the agent role over the default (causer) option. Let’s take another look at the contrast in (4). Example (4a) illustrates the default interpretation, in which the subject is conceived as a causer. The anchor event is therefore identified with the change (the showing event that takes place within the indirect object’s referent, namely their discovery of the problem). Its occurrence cannot therefore be denied in the subsequent discourse without contradiction.

- (4) a. This situation showed them the problem, #but they didn’t see it.’
- b. Clearly, this situation objectively showed them the problem! It is crazy that they didn’t see it!

But in (4b), the adverbs *clearly* and *objectively* instruct us to focus on what the subject’s referent *does* in the anchor event (e.g. emitting signals that indicate the problem), and to *ignore all other additional circumstances*. The indirect object’s referent can then perceive these signals in normal possible extensions of the anchor event only.¹⁴

This type of rescue is only possible if the action of the inanimate considered apart from any other circumstance can be indicative of the outcome *before* the outcome takes place (see section 4.2.2). This is typically not the case in (12a/b): the impact on the watch can most likely only be identified as the cause of the repair once the watch is fixed. This explains why the statements in (12a/b) sound rather nonsensical.¹⁵ The retrospective inferential path from result to cause is also most likely at play in (11), as fate is most often invoked as a causal factor *post facto*, in an attempt to give events that have already happened coherence and causal structure, or to endow them with a fatalistic inevitability. This accounts for why (11) tends to trigger a result entailment regardless of the modal flavour at play.

5 The ingredients of causal defeasibility

Modal causatives raise more questions than can fully be addressed in this chapter. One of these is: What properties must a causative verb satisfy in order to become defeasible? Without claiming to be exhaustive, I can offer three points in response.

The first concerns the *perceptual inaccessibility* of the targeted outcome. Most defeasible causatives discussed in Martin & Schäfer (2012) have an animate direct or indirect object: teachers, showees, and so on are all animates.¹⁶ Relatedly, most of them denote actions whose relevant effects are not directly observable: rather than producing an overt physical transformation, they alter the theme’s psychological or epistemic state, or their social/normative relation. Since such changes are largely unobservable, the successful and unsuccessful variants of these actions are typically indistinguishable on the surface. This certainly makes it easier to disregard whether the outcome has occurred or not when determining whether the sentence is true. Compare this with run-of-the-mill causative predicates like *open a box of sweets*. Such causative predicates describe outcomes that cannot escape senses. This reinforces the requirement that the change must take place for the sentence to be considered true.

¹⁴Note that DPs like *the situation* can pass the *do* agentivity test (see e.g. *What this situation did was shake up the family dynamic*), which suggests that the referent of such DPs can be conceived as inanimate agents.

¹⁵We also understand the reviewer’s intuition reported in footnote 5: in their scenario where a realignment of a mechanical component should normally suffice to make the watch functional again, it is possible to foresee the outcome from the shock alone, before the outcome actually takes place.

¹⁶48 out of 55 verbs from Martin & Schäfer’s (2012) list have an animate direct or indirect object.

The second point concerns the *extensional identity* between the successful and non-successful variants of the actions described. With well established defeasible causative verbs, the action normally carried out to trigger the target outcome can be performed in exactly the same way when it is unsuccessful as when it is successful. Recall Ryle’s (1949) examples of repairs and medical treatments. Or take *open*. On a computer, we can do exactly what we would do to open a file, but without the file actually opening. Conversely, we cannot do exactly what is needed to do to open a box of sweets without actually opening the box. And it seems that we can say: *But I’ve opened the file! Why isn’t it open?* whereas we cannot say *But I’ve opened the box of sweets! #Why isn’t it open?* It seems that the outcome tends to be entailed when the anchor event must contain at least a part of the change for it to be indicative enough of the right type of causation.

A last point concerns the information conveyed by the verb about the *manner* or *way of causing*. As mentioned above, defeasible causative verbs have a property which is rather uncommon for causative verbs in the traditional definition: they encode an event (manner) property (they say something about the activity that causes the result). In fact, some of the French lexical causatives which were defeasible in the past before becoming undefeasible (or ‘neutral’) also encode manner information. For instance, *assassiner* ‘assassinate’ implies but does not entail death until the XIX century, and therefore had failed attempt readings like in (25) (example from Vogeleer & Guéron 2022, 17; see also Koenig et al. 2017). But this verb also says something about the way the subject’s referent is acting (i.e. intentionally, by a sudden or secret attack). Encoding a manner of causing might be a necessary condition for the action *alone* to be sufficiently indicative of the right causation type.

- (25) *On assassina Luc, et on l’a manqué.*
 one assassinated Luc and one him has missed
 ‘Someone assassinated Luc and they missed him.’ (Voltaire, Lettre à d’Alembert, 1762)

Voltaire could not have used *tuer* ‘kill’ instead of *assassiner* in (25), as *tuer* is result-entailing also at Voltaire’s time. It is tempting to link this to the fact that *tuer* does not specify anything about the activity causing the result.

The coexistence of neutral and defeasible causative verbs in the lexicon undoubtedly presents a challenge for the child, who must therefore learn to identify the ‘exact size’ of the anchor event for each causative verb (for instance, an action suffices for *send the box of sweets* to be true, but not for *open the box of sweets* to be true). This could explain the non-target uses of these verbs reported in the literature. For instance, Tomasello (1992) reports “packaging errors” in development in that children tend first to interpret certain causative verbs like *fix* or *inflate* as describing mere actions (e.g., trying to repair something, blowing in a balloon) rather than the causation (action + change-of-state) meaning that these verbs encode; see also Wittek (2002) for a similar pattern in child German. However, other studies, such as those by Kazanina, Baker & Seddon (2020), report an opposite pattern in verb learning. Their experimental results show that L1 English learners understand sentences such as *Mary sent a book to John* as entailing successful transfer rather than implying it. Such non-target like patterns are not surprising, given that the size of the anchor of causative verbs varies considerably from one verb to another, and from one use to another.

Acknowledgments

I would like to thank Florian Schäfer for invaluable comments on previous versions of this chapter, and the anonymous reviewer for their constructive and detailed feedback. Thanks also to Despina Oikonomou and the audience of OASIS 5 in Edinburgh for discussion.

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